

Modules 1–3: Introduction to Data Analytics › Statistics & Probability › Advanced Excel for Analytics

PROGRAM	ASSESSMENT CODE	TOTAL TIME	STACK
Data Analytics	M3-A1	4 Hours	Microsoft Excel / Python 3

GENERAL INSTRUCTIONS

Attempt all sections. For Excel tasks, submit the .xlsx file with all formulas, pivot structures, and formatting intact — rename it YourName_M3-A1.xlsx. For Python tasks, submit your .py or .ipynb file along with a screenshot of the console or notebook output. Label every task clearly with its section and task number. Do not use external data sources unless a task explicitly instructs you to. All work must be your own.

A

Section A — Concept Application

Answer all questions. Each question presents a real-world food delivery scenario followed by a concept-specific question requiring reasoning, comparison, or justification — not recall. Write 4–6 lines per answer. Questions 1–2 draw from Module 1, Questions 3–4 from Module 2, and Questions 5–6 from Module 3.

SCENARIO

S
1

You are a junior data analyst at a food delivery platform. Your manager hands you a raw dump of 80,000 order records and asks for three separate reports: (a) a summary of what happened last month — total orders, revenue, and cancellation rate; (b) a diagnosis of why delivery delays spiked by 40% in Week 3; and (c) a forecast of expected order volumes for the next quarter based on historical trends.

Question: Identify which type of analytics — Descriptive, Diagnostic, Predictive, or Prescriptive — applies to each of your manager's three requests. Explain in your own words why each report belongs to that category, what inputs each analysis requires, and what kind of output or decision it should produce.

SCENARIO

S
2

You are building a business health report for the operations head of a food delivery startup. She asks you to track two things: 'How healthy is our business right now?' and 'How well are we retaining customers over time?' You must select and define the right KPIs for each question from the available order and customer data.

Question: Define at least two KPIs for each question. For each KPI, write the exact calculation formula and explain how an upward or downward movement in that KPI signals a specific business problem or improvement. Why is domain understanding critical when choosing KPIs over simply reporting raw counts or averages?

SCENARIO

S
3

You are analysing delivery time data for a food delivery platform. The dataset has 12,000 rows. Your descriptive summary shows: Mean delivery time = 38 minutes, Median = 29 minutes, Standard Deviation = 24 minutes. An investigation reveals that 60 orders logged delivery times of over 200 minutes due to a GPS tracking error on a public holiday.

Question: Based on these statistics, explain what the gap between the mean and median tells you about the shape of the delivery time distribution. Which measure of central tendency should you report to the operations team to represent typical delivery performance, and why? What does the high standard deviation indicate, and how would you decide whether to remove, cap, or keep the outlier deliveries before presenting findings to management?

SCENARIO

S
4

You are a data analyst at a food delivery company. The marketing team ran a 'Free Delivery Weekend' campaign and claims it significantly increased average order value. You have two weeks of order data before the campaign and two weeks after. Your manager asks you to statistically validate this claim before the company commits budget to running it again.

Question: Which statistical test would you use to validate the marketing team's claim, and why is it the appropriate choice for this scenario? State the null and alternative hypotheses clearly. If your test returns a p-value of 0.03 at a 95% confidence level, what decision do you make, and how would you communicate the result in plain business language to the marketing team?

S
5

SCENARIO

You are working on an Excel reporting workbook for a food delivery company. The workbook has three sheets: Orders (100,000 rows: OrderID, CustomerID, RestaurantID, Cuisine, OrderDate, DeliveryTime, Rating, Revenue), Restaurants (RestaurantID, Name, Zone, Category), and Customers (CustomerID, Name, Tier, City). Your manager asks you to pull RestaurantName and Zone into the Orders sheet so the analyst team can filter and summarise by those fields.

Question: Explain why VLOOKUP would be problematic for this specific task and how XLOOKUP or INDEX-MATCH resolves those limitations. Write the Excel formula you would use to return the RestaurantName into the Orders sheet and describe what each argument does. What would happen if a RestaurantID in Orders has no match in the Restaurants sheet, and how would you handle that situation in your formula?

S
6

SCENARIO

You are designing a monthly performance dashboard in Excel for the operations director of a food delivery company. The dashboard must display three things: total revenue broken down by cuisine category, weekly order volume trends over the past quarter, and the top 5 delivery zones ranked by average customer rating — all of it updatable with a single click when new monthly data is appended.

Question: Describe the specific Excel features — Pivot Tables, Pivot Charts, Slicers, Timelines, and Conditional Formatting — you would use to build each of the three dashboard components. Explain how Slicers and Timelines connect to Pivot Tables to make the dashboard interactive. Why is it critical to structure the source data as a formal Excel Table (Ctrl+T) before building these components, and what breaks in the dashboard if you skip that step?

B

Section B — Practical Tasks

Complete all tasks. Tasks 1–3 use Microsoft Excel. Task 4 uses both Excel and Python 3 (pandas). Submit all .xlsx and .py / .ipynb files, each labelled with your name and task number. Requirements within each task are mandatory — partial submissions will not be graded.

Task 1: Food Order Data Cleaning & Conditional Views

Build a clean, filtered, and conditionally formatted view of a food delivery orders dataset using Excel's core data tools.

- Create a dataset of at least 30 rows with columns: OrderID, CustomerName, Cuisine, OrderDate, DeliveryTime (minutes), Rating (1–5), Revenue (Rs), and Status (Delivered / Cancelled / Delayed).

- Apply Data Validation to the Cuisine column using a dropdown list of exactly 5 cuisine types, and to the Rating column to restrict entries to whole numbers between 1 and 5 with an input message and error alert.
- Use Advanced Filter with a criteria range to extract all rows where Status = 'Delivered' AND Revenue > Rs 500 AND Rating >= 4 into a separate sheet named 'Premium Orders'.
- Apply Conditional Formatting to the DeliveryTime column using a colour scale: green for <= 30 minutes, amber for 31–45 minutes, and red for > 45 minutes. Use Icon Sets or a three-colour scale rule.

Task 2: Restaurant & Zone Lookup Enrichment

Build a multi-sheet lookup system that enriches an orders table with restaurant and delivery zone details using XLOOKUP and INDEX-MATCH.

- Create a Restaurants reference sheet with at least 15 rows and columns: RestaurantID, RestaurantName, Zone, Category (Cloud Kitchen / Dine-in / QSR), and AvgRating.
- In the Orders sheet, use XLOOKUP to pull RestaurantName from the Restaurants sheet based on RestaurantID. Include the not-found argument so unmatched IDs display 'Not Listed' instead of an error.
- Use INDEX-MATCH to pull the Zone value for each order into a separate column — demonstrate the two-function nested structure clearly.
- Add a summary table below the orders data using AVERAGEIF to calculate the average Revenue for each unique Zone, pulling zone names dynamically from the Restaurants sheet.

Task 3: Pivot Revenue & Trend Dashboard

Build a multi-layer Pivot Table analysis of food delivery revenue with date grouping, a calculated field, slicers, and a linked Pivot Chart.

- Create an Excel Table (named tblOrders) from a dataset of at least 50 rows containing: OrderDate, Cuisine, Zone, Revenue, DeliveryTime, and Status columns.
- Build Pivot Table 1 showing Revenue (SUM) by Cuisine (rows) × Zone (columns), with Status as a Report Filter. Include Grand Totals for both rows and columns.
- Build Pivot Table 2 showing Order Count by Month and Quarter, with the OrderDate field grouped. Add a Timeline slicer connected to this pivot for date filtering.
- Add a Calculated Field named AvgRevenuePerOrder to Pivot Table 1 using the expression = Revenue / Orders (where Orders is the count field). Display it alongside the standard SUM(Revenue) column.
- Insert a clustered bar Pivot Chart linked to Pivot Table 1. Add a Slicer for Cuisine that controls both the Pivot Table and the Pivot Chart simultaneously.

Task 4: Delivery Performance EDA — Excel + Python

Perform a structured exploratory data analysis of food delivery performance metrics using Excel formulas for descriptive statistics and a Python pandas script for correlation and outlier detection.

- In Excel, use dedicated functions to calculate the following statistics for the DeliveryTime column: Mean (AVERAGE), Median (MEDIAN), Mode (MODE.SNGL), Standard Deviation (STDEV.S), Variance (VAR.S), Min, Max, and the 25th and 75th percentiles (QUARTILE.INC). Present them in a labelled summary table on a separate sheet named 'Stats Summary'.
- Still in Excel, apply the IQR method to flag outliers in the DeliveryTime column: compute $IQR = Q3 - Q1$, define lower bound = $Q1 - 1.5 \times IQR$ and upper bound = $Q3 + 1.5 \times IQR$ using an IF formula, and add a new column named 'Outlier Flag' that displays 'Yes' for out-of-range values and 'No' otherwise.
- In Python (pandas), load the same dataset from a CSV export of the Excel table. Replicate the descriptive statistics using .describe() and manual IQR calculation. Print a clean, labelled summary to the console.
- In Python, compute a correlation matrix between Revenue, DeliveryTime, and Rating using .corr(). Print the matrix and write at least 3 lines of inline comments in the code explaining what each correlation coefficient implies for the business — for example, what a negative correlation between DeliveryTime and Rating means for operations.

C

Section C — Mini Capstone Project

Mini Project: Food Delivery Operations Dashboard — Excel Edition

Objective:

Build a complete, interactive operations dashboard for a fictional food delivery company in a single Excel workbook, combining lookup enrichment, pivot analysis, statistical summaries, and conditional formatting into one connected reporting system that an operations manager can use without touching raw data.

Your project must:

- **Source Data:** Simulate or create a dataset of at least 100 orders across 5 delivery zones, 4 cuisine types, and 3 months. Structure it as a proper Excel Table named tblOrders with columns: OrderID, RestaurantID, Cuisine, Zone, OrderDate, DeliveryTime, Rating, Revenue, and Status.
- **Data Enrichment:** On a separate Restaurants sheet, maintain at least 15 restaurant records. Use XLOOKUP to pull RestaurantName and Category into the Orders table for every row; all lookups must include error handling so missing IDs display 'Unknown' rather than #N/A.
- **Statistical Summary Sheet:** Create a sheet named 'Zone Stats' that shows Mean, Median, and Standard Deviation of DeliveryTime and Revenue for each of the 5 zones, calculated using AVERAGEIF and array-based STDEV formulas. Also flag which zones have an average DeliveryTime above the overall median using a conditional formula.

- **Interactive Pivot Dashboard:** On a sheet named 'Dashboard', build at least 2 Pivot Tables — (a) Revenue by Cuisine × Zone and (b) Order Count by Month — linked to the tblOrders table. Add a shared Zone Slicer that controls both Pivot Tables and a Timeline Slicer for date-range filtering.
- **KPI Cards & Conditional Formatting:** Above the pivots, display 3 dynamic KPI summary cells: Total Revenue (SUM of Revenue), Average Customer Rating (AVERAGE of Rating), and On-Time Delivery % (count of rows where Status = 'Delivered' AND DeliveryTime ≤ 40 minutes, divided by total rows). Apply a Top 10% conditional formatting rule to the Revenue column in tblOrders to highlight high-value orders in green.

D

Section D — AI-Augmented Learning

STEP 1 • BUILD WITH AI

Use an AI tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to help you write a Python script (pandas) that:

- Reads a food delivery orders CSV file and produces a summary table showing Total Revenue, Average DeliveryTime, and Order Count grouped by Cuisine.
- Adds a new column named PerformanceFlag that labels any row with DeliveryTime > 60 as 'Delayed' and all others as 'On Time'.
- Saves the cleaned and enriched output as a new file named food_delivery_summary.xlsx using pandas ExcelWriter.
- Runs without raising an error on a dataset where some Revenue cells are blank and at least one DeliveryTime value is negative (a data entry error).

STEP 2 • TEST & DEBUG (WITHOUT AI)

Then, working without AI, test the generated script against a dataset that includes at least 3 blank Revenue cells and one DeliveryTime of -5. Find at least one bug or gap in the AI's solution — for example, how it handles NaN revenue in aggregation, whether it catches negative delivery times, or how it behaves on completely empty rows — and fix it yourself.

SUBMIT

All 3 items required

- 1 The exact prompt(s) you gave the AI tool.
- 2 The AI's original script and your corrected version, both clearly labelled in separate files or clearly marked sections.
- 3 A 3–4 line written note explaining what you changed, which specific input exposed the bug, and why the AI's original version failed on that input.